Week 3 · Day 2 · Subnets — Public vs Private

Configure Route Tables & Internet Gateway

⏱ Time	65 min
Objective	Attach an Internet Gateway to your custom VPC, create public and private route tables, associate subnets correctly, and verify the routing logic that makes subnets truly public or private.

Prerequisites — Complete Before Starting

Custom VPC from Week 3 Day 1

Portfolio Goal — What You Will Post

Add a traffic flow diagram to your portfolio showing how packets travel through your VPC — this demonstrates deep networking understanding.

Lab Steps

Step 1 Create and Attach an Internet Gateway

1. Go to VPC → Internet gateways → Create internet gateway.
2. Name tag: OluTech-IGW.
3. Click Create internet gateway.
4. After creation, click Actions → Attach to VPC.
5. Select OluTech-Production-VPC. Click Attach internet gateway.
6. Status should now show: Attached.

Step 2 Create a Public Route Table

1. Go to VPC → Route tables → Create route table.
2. Name: Public-Route-Table.
3. VPC: OluTech-Production-VPC. Click Create.
4. Select your new route table → Routes tab → Edit routes.
5. The default local route (10.0.0.0/16 → local) is already there — do NOT delete it.
6. Click Add route: Destination: 0.0.0.0/0 | Target: Internet Gateway → select OluTech-IGW.
7. Click Save changes.
8. This 0.0.0.0/0 → IGW route is what makes subnets PUBLIC.

Step 3 Associate Public Subnets with Public Route Table

1. Select Public-Route-Table → Subnet associations tab → Edit subnet associations.
2. Check both: Public-Subnet-AZ-A and Public-Subnet-AZ-B.
3. Click Save associations.

Step 4 Verify Private Subnets Have No Internet Route

1. Check the main/default route table for your VPC — it should have only the local route (10.0.0.0/16 → local).
2. The two private subnets should be associated with this main route table (no IGW route).
3. This confirms: your private subnets have NO path to the internet — correct for databases and app servers.

Step 5 Draw the Traffic Flow Diagram

1. In Excalidraw, draw arrows showing how traffic flows:
2. Internet User → Internet Gateway → Public Subnet → EC2 Web Server.
3. EC2 Web Server → Private Subnet (via local route) → EC2 App Server → RDS Database.
4. Show that Private Subnets have no arrow to the Internet Gateway.
5. Label each arrow with the route table entry that enables it.
6. Export as PNG.

📸 Screenshot Checklist

Take these screenshots and save them to your portfolio folder:

- ☐ Internet Gateway attached to your VPC (Attached status)
- ☐ Public Route Table showing local route + 0.0.0.0/0 → IGW
- ☐ Public subnets associated with Public Route Table
- ☐ Private subnets associated with main route table (local route only)
- ☐ Your traffic flow diagram

📱 Portfolio Post — Copy & Customise

LinkedIn: 'Deep-dived into VPC routing today. The single most important thing I learned: a subnet is only PUBLIC if its route table has a 0.0.0.0/0 route pointing to an Internet Gateway. The name means nothing — the routes do. Traffic flow diagram attached. #VPC #Networking #AWSarchitecture'

🔥 Bonus Challenge — Push Your Skills Further

Create a NAT Gateway in one of your public subnets (note: ~$0.045/hour — delete it after 15 minutes to stay free tier). Add a route in the private route table: 0.0.0.0/0 → NAT Gateway. Explain: how does this differ from the IGW route? What does this enable?

✓ Lab Complete — Mark Off Your Progress

Lab Steps Completed	☐
All Screenshots Taken	☐
Portfolio Artifact Posted	☐
Bonus Challenge Attempted	☐